

BUS13-3

Financial Accounting & State- ments

Reading the three statements, ratios, and working capital from the
user's perspective

Financial statements are constructed documents shaped by accounting choices and
management incentives.

Albert School — 22 hours

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1 What accounting measures vs. what finance cares about

Accounting produces a periodic record of a firm's transactions according to a set of conventions. Its aim is to measure, for a given period, what the firm **owns** and **owes** (the balance sheet) and what it **earned** (the income statement), following rules of recognition and valuation that prioritize verifiability and consistency over economic relevance.

Finance cares about something different: the **cash** a firm will generate in the future and the **risk** attached to it. Accounting figures are a starting point, but several conventions make book numbers diverge from economic reality:

- **Historical cost.** Assets are often carried at what was paid for them, not what they are now worth. A building bought decades ago sits on the books at a value unrelated to its current market price.
- **Accrual recognition.** Revenue and expenses are booked when earned or incurred, not when cash moves. Reported profit is therefore not cash.
- **Off-balance-sheet items and intangibles.** Internally generated brands, human capital, and many commitments are not recognized as assets, so book value omits real sources of value.

These three are the standard examples where book values mislead, and they recur throughout this course.

2 The balance sheet

The balance sheet states, at a point in time, the firm's resources and how they are financed. It obeys the identity

$$\text{Assets} = \text{Liabilities} + \text{Equity}.$$

2.1 Assets

Assets are split by horizon:

- **Non-current (fixed) assets:** held beyond one year — property, plant and equipment, intangibles, long-term investments.
- **Current assets:** expected to convert to cash within the operating cycle or one year — inventories, receivables, cash.

Assets are valued either at **historical cost** (acquisition price, possibly less depreciation) or at **fair value** (an estimate of current market value). The two differ because historical cost ignores subsequent price changes, while fair value tracks them. The choice affects both reported asset values and profit.

2.2 Liabilities

Liabilities are obligations to third parties, also split into non-current (long-term debt, provisions) and current (payables, short-term borrowings, accruals).

2.3 Equity

Equity is the residual: $\text{Equity} = \text{Assets} - \text{Liabilities}$. It comprises share capital, reserves, and retained earnings.

Definition 1 **Book equity** is the accounting residual above. **Market equity** (market capitalization) is the price investors place on the firm's shares.

Book equity rarely equals market equity because book equity is built from historical-cost assets and excludes unrecognized intangibles and future growth, whereas market equity reflects investors' expectations of future cash flows and risk.

3 The income statement

The income statement reports performance over a period. Revenue and expenses are recognized on an accrual basis.

3.1 Revenue recognition

Definition 2 **Revenue recognition** is the principle determining **when** a sale is recorded: broadly, when control of the good or service has transferred to the customer and the amount is measurable, not necessarily when cash is received.

3.2 Operating vs. financing expenses

Expenses are separated into **operating** (costs of running the business: cost of goods sold, wages, depreciation) and **financing** (interest on debt). The separation isolates operational performance from the effect of how the firm is financed.

3.3 Margin levels

Reading downward, the statement yields successive measures of profit:

- **EBITDA** — earnings before interest, taxes, depreciation and amortization;
- **EBIT** (operating profit) — EBITDA less depreciation and amortization;
- **EBT** — EBIT less net interest;
- **Net income** — EBT less taxes.

Definition 3

$$\text{EBITDA} = \text{EBIT} + \text{depreciation} + \text{amortization}.$$

Each level answers a different question. EBITDA approximates operating cash generation before capital intensity; EBIT measures operating profitability after the cost of using fixed assets; net income is what accrues to shareholders after financing and tax. Which one matters depends on the comparison: EBIT and EBITDA for operational efficiency independent of capital structure, net income for returns to equity.

4 The cash flow statement

The cash flow statement explains the change in cash over the period across three activities.

4.1 Operating activities (indirect method)

The **indirect method** starts from net income and adjusts it to operating cash flow:

1. add back non-cash charges (depreciation, amortization, provisions);
2. remove non-operating items;
3. adjust for changes in working capital (increases in receivables and inventories consume cash; increases in payables release cash).

This reconciliation is the reason net income $\neq$ cash flow: accruals, non-cash charges, and working-capital movements all drive the two apart. A profitable firm can be cash-negative when, for instance, it is growing receivables and inventory faster than it collects.

4.2 Investing activities

Cash spent on or received from acquiring and disposing of long-term assets — principally capital expenditure (capex).

4.3 Financing activities

Cash from and to providers of capital — issuing or repaying debt, issuing shares, paying dividends.

4.4 Cash flow patterns by lifecycle

The signs of the three cash flows signal where a firm sits in its lifecycle:

- **Startup / growth:** operating cash flow negative or weak, investing negative (heavy capex), financing positive (raising capital);
- **Mature:** operating positive, investing negative but modest, financing negative (repaying debt, paying dividends);
- **Decline:** operating falling, investing possibly positive (selling assets).

5 How the three statements connect

The statements are not independent; they articulate:

- **Net income** from the income statement is the starting line of operating cash flow and feeds **retained earnings** in equity on the balance sheet.
- **Capex** in the investing section adds to the **fixed-asset** base on the balance sheet and generates future **depreciation** in the income statement.
- **Financing** flows change **debt and equity** on the balance sheet, i.e. the capital structure.

- The closing **cash** balance from the cash flow statement is the cash line on the balance sheet.

Worked example. A sale on credit is recognized as **revenue** (income statement), raising net income; it creates a **receivable** (balance-sheet asset) rather than cash; the cash flow statement removes the uncollected amount as an **increase in working capital**, so operating cash flow rises by less than net income. When the customer later pays, the receivable falls and cash rises, with no further effect on income.

6 Earnings quality

Definition 4 **Earnings quality** is the degree to which reported earnings reflect sustainable, repeatable economic performance rather than one-off events or accounting discretion.

Assessing it requires:

- **Identifying non-recurring items:** restructuring charges, asset write-downs, gains on disposals, litigation settlements — items unlikely to repeat.
- **Normalizing earnings:** removing those one-time items to obtain a figure representative of ongoing operations.
- **Flagging red flags of manipulation:** profit rising while operating cash flow stalls, receivables or inventory growing faster than sales, frequent “one-time” charges, changes in accounting estimates that conveniently lift earnings.

Aggressive **revenue recognition** is a common lever: **channel stuffing** (pushing goods onto distributors to book sales early), front-loading **long-term contracts**, and timing of **subscription** revenue can all inflate reported earnings ahead of the underlying economics.

7 Financial ratios: the four families

Ratios summarize the statements into comparable indicators. Four families matter here.

Definition 5 Liquidity — ability to meet short-term obligations.

$$\text{Current ratio} = \frac{\text{current assets}}{\text{current liabilities}}, \quad \text{Quick ratio} = \frac{\text{current assets} - \text{inventories}}{\text{current liabilities}}.$$

Definition 6 Solvency — ability to meet all obligations / reliance on debt.

$$\text{Debt-to-equity} = \frac{\text{total debt}}{\text{equity}}.$$

Definition 7 Profitability — ability to generate profit from sales.

$$\text{Gross margin} = \frac{\text{gross profit}}{\text{revenue}}, \quad \text{Net margin} = \frac{\text{net income}}{\text{revenue}}.$$

Definition 8 Efficiency — how well assets are used to generate sales.

$$\text{Asset turnover} = \frac{\text{revenue}}{\text{total assets}}.$$

Interpretation: a higher current or quick ratio signals more comfortable short-term liquidity; a higher debt-to-equity signals greater financial risk; a higher gross or net margin signals stronger profitability; a higher asset turnover signals more sales extracted per unit of assets. Each is read relative to the firm's history and its industry peers, not against a universal threshold.

8 Working Capital Requirement (BFR)

Definition 9 The **Working Capital Requirement** (Besoin en Fonds de Roulement, BFR) is the operating cash tied up in the cycle between paying suppliers and collecting from customers:

$$\text{BFR} = (\text{inventories} + \text{trade receivables}) - \text{trade payables}.$$

The drivers are the operating cycle itself: how long inventory is held, how long customers take to pay, and how long the firm takes to pay suppliers. A positive BFR means the operating cycle **consumes** cash that must be financed; a negative BFR (customers pay before suppliers are paid, as in some retail) means the cycle **releases** cash. Reading BFR from the balance sheet therefore reveals how much cash a company's day-to-day operations absorb or free up, independent of its reported profit.

9 Consolidated accounts and consolidation methods

When a company controls or influences others, **consolidated accounts** present the group as a single economic entity. The method depends on the degree of control:

- **Full consolidation** — for **control** (typically majority of voting rights): 100% of the subsidiary's assets, liabilities, revenues and expenses are combined, with a **minority (non-controlling) interest** line for the share not owned by the parent.
- **Proportional consolidation** — for **joint control**: the investor's percentage share of each line is combined.
- **Equity method** — for **significant influence** (typically 20–50%): the investment appears as a single line on the balance sheet, increased by the investor's share of the associate's profit.

10 French GAAP vs. IFRS

French GAAP (the Plan Comptable Général) and IFRS differ in ways that affect cross-border comparison:

- **Valuation basis**: French GAAP leans on historical cost and prudence; IFRS makes wider use of fair value (e.g. for financial instruments and investment property).
- **Orientation**: French GAAP is influenced by legal and tax requirements; IFRS is oriented toward investors and economic substance over legal form.
- **Specific treatments**: differences in the treatment of development costs, leases, goodwill, and provisions change reported assets and earnings.

The implication is that two otherwise identical firms reporting under different standards can show different book values and profits, so analysts must adjust for the standard before comparing companies across jurisdictions.

11 Reading a real annual report end-to-end

Bringing the course together, a first-pass analysis of a company's annual report proceeds as follows:

1. **Identify the business model** — what the company sells, to whom, and how it makes money.
2. **Compute the key ratios** across the four families and the **BFR**.
3. **Flag financial signals** — at least three observations from the statements and ratios (e.g. a margin trend, a liquidity concern, a cash-flow vs. profit gap).
4. **Raise structural or governance questions** — at least three matters warranting deeper investigation (e.g. earnings-quality concerns, related-party dealings, off-balance-sheet exposures).

The output is a structured, numbers-first assessment of the company's health, not a narrative summary.

What to retain. The three statements are constructed documents: the balance sheet shows position, the income statement performance, the cash flow statement the movement of cash — linked through net income, capex, financing, and the cash balance. Ratios (liquidity, solvency, profitability, efficiency) and BFR summarize health in numbers; earnings quality and the choice of accounting standard tell you how far to trust those numbers.